

ArogyaRakshak — Answers to the 16 Identified Challenges

For each point: the honest answer, then the concrete fix to adopt so the project stays viable and truthful (not oversold).

Module 1 — BillNyay

1. Document Authenticity

Honest answer: No, the system cannot verify authenticity. You have no hospital/insurer database access, and OCR reads text — it doesn't detect forgery. This is a real, permanent limitation, not something you engineer away in an FYP timeline.

Fix — don't solve authenticity, add integrity checks that catch *sloppy* fakes without claiming to catch *good* ones:

- **Arithmetic consistency check:** do line items actually sum to the stated total? Mismatches are cheap to flag and catch a real class of errors/edits.
- **Internal consistency check:** date logic (admission before discharge), duplicate line items, formatting anomalies vs. known hospital bill templates.
- **Duplicate submission detection:** hash uploaded documents; flag if the same document (or a near-duplicate) was submitted before under a different case.
- **Explicit disclaimer** on every output: "This analysis assumes the uploaded document is genuine. ArogyaRakshak does not verify document authenticity." This turns an unanswerable question into a scoped, correctly-labeled feature — which is the strongest possible answer in a viva.

2. CGHS Benchmark Limitation

Honest answer: Correct, private hospitals aren't bound to CGHS rates, and your documentation already gets this right — "flag deviation," never "declare overcharging." Keep that language discipline everywhere (UI copy, PDF output, report).

Fix — make the benchmark itself smarter instead of retreating from it:

- CGHS rates are already published with **city-tier categorization** (e.g., different rates for metro vs. non-metro). Ingest that tiering instead of one flat rate table — this alone meaningfully improves fidelity and is a real, gradeable technical improvement.
- Where your data has it, adjust for room category/type (general vs. private ward) since CGHS schedules differentiate these too.
- Frame the output as "**X% above the CGHS reference benchmark for [city tier / room category]**" — specific, defensible, and honest about being a benchmark, not a verdict.

3. Actionability

Honest answer: Correct that a comparison alone doesn't solve the user's problem — but your current scope already goes past "just showing a comparison": review report + drafted grievance + document checklist + next steps. That's meaningfully actionable for a document-based system with no insurer integration.

Fix — close the loop slightly further without needing insurer API access:

- Add a simple **status tracker**: user self-reports what happened after they filed ("submitted," "response received," "resolved") — gives you real outcome data for your final report/eval, and costs almost nothing to build.
- Add **reminder nudges** (email/in-app) for follow-up deadlines mentioned in the appeal checklist.
- Explicitly scope direct insurer/TPA API submission as **future work**, not something the FYP needs — that's an honest, correct boundary to state.

4. Legal Responsibility

Honest answer: No, and you shouldn't try to. This is correctly scoped already — informational benchmarking, final decision with the concerned authority.

Fix — make the boundary airtight in output, not just in your head:

- Never use the word "overcharged" as a system verdict — use "flagged for review" or "deviates from benchmark by X%" consistently across UI, PDF, and appeal letter templates.
- Add a **disclaimer clause inside the generated appeal letter itself**, not just the UI — something like "This document was prepared using an automated benchmarking tool and is provided as a starting point; the user is responsible for reviewing before submission."
- Worth getting your guide (or a legal-studies contact at your college, if any) to glance at the disclaimer wording once — cheap insurance for viva.

Module 2 — SchemeSetu

5. Scheme Rule Changes

Honest answer: Yes, real risk — PMJAY/MJPJAY rules do change. Your "periodic RAG update" scope is the right approach; the gap is that "periodically" isn't currently a concrete process.

Fix:

- Assign this explicitly to Role 5 (data + legal/eval) as a **recurring task with a cadence** — e.g., check PMJAY/MJPJAY portals monthly, log any circular updates, re-ingest into

the RAG store.

- Display a "**data last verified: [date]**" timestamp on every SchemeSetu output — this is honest, cheap, and actually a good viva answer ("we surface data freshness to the user rather than pretending our benchmark is always current").

6. Eligibility vs Approval

Honest answer: Correct, no guarantee is possible — actual approval is a government verification step you have no access to.

Fix: Already scoped correctly (estimate + guidance only). Just make sure the UI language matches: "Estimated eligibility" as a label, never "You qualify for X," and include a confidence indicator if the RAG retrieval was ambiguous. This is a labeling discipline problem, not a technical one — cheap to fix, easy to get wrong if you don't set the copy conventions early.

Module 3 — DawaCheck

7. Premium Medicines

Honest answer: Correct — price $\neq$ overpricing for branded/patented/non-scheduled drugs, and most premium medicines sit outside NPPA regulation entirely.

Fix: Already correctly scoped to NPPA Schedule-I only, which is the one segment with a clean government ground truth. Keep it there. Resist any team pressure to "expand coverage" to non-scheduled drugs mid-build — there's no defensible benchmark to check against, and it would undermine the credibility of the whole module. This is a case where the right fix is discipline, not more engineering.

8. Trust in Generic Alternatives

Honest answer: Correct concern — a patient shouldn't trust an AI over their doctor's prescription, and your scope already avoids recommending a switch.

Fix — make it actionable without overstepping:

- Frame purely as "your prescribed medicine has a generic equivalent available at [Jan Aushadhi / PMBJP store locator]" — informational, points to an official government resource, doesn't tell the user what to do.
- Explicit copy: "Discuss with your doctor or pharmacist before switching any medication."
- This keeps the feature useful (awareness) while staying strictly out of clinical decision-making.

9. User Behaviour

Honest answer: You're right that you can't control whether people act on cost information

— that's outside the system's job.

Fix: This isn't a bug to fix, it's a metric to *measure* honestly in your evaluation. Track (opt-in) whether users viewed the generic-alternative suggestion, not whether they switched medicines — "awareness delivered" is a legitimate, measurable outcome for your report; "behavior changed" was never a fair success metric for this module and you shouldn't claim it.

Cross-Module Challenges

10. Accuracy of OCR

Honest answer: Real limitation, correctly scoped (show confidence, request better image if low).

Fix — add one more layer: a lightweight **manual correction step** — after OCR extraction, show the user the extracted fields (amount, date, medicine name) and let them correct obvious errors before the pipeline proceeds. This is standard practice in production document-AI products, cheap to build, and turns "OCR might be wrong" from a weakness into "OCR is a draft, human-in-the-loop confirms it" — a much stronger viva answer.

11. AI Hallucinations

Honest answer: Real risk with any LLM, mitigated (not eliminated) by RAG grounding.

Fix — go one step further than "restrict responses to retrieved information":

- Have the QA Judge agent (already in your BillNyay pipeline) explicitly **cite which retrieved document/clause** supports each claim in the generated appeal — if it can't cite a source, that claim gets dropped or flagged, not included.
- Surface those citations to the user too ("per CGHS Circular X, dated Y") — this is good practice and doubles as evidence for your report that hallucination mitigation is a designed control, not an afterthought.

12. Multilingual Accuracy

Honest answer: Real risk — translation/terminology errors could genuinely affect medical/financial meaning, which is higher-stakes than typical MT error.

Fix: Already scoped correctly (IndicSBERT/IndicXlit + QA testing). Make sure Role 5's "multilingual QA" isn't a one-time pass — build a small **labeled test set of medical terms/phrases in all three languages early**, and re-run it every time you touch the entity resolution or generation pipeline. This is also exactly the labeled dataset your Kadi entity-resolution ablation study (for the IndiaAI fellowship note) already needs — same work serves both purposes.

13. Data Freshness

Honest answer: Real, ongoing problem — government pricing/scheme data changes on its own schedule, not yours.

Fix: Same pattern as #5 — versioned datasets (already scoped) + a **visible "last updated" date** on every output that used reference data. Assign an explicit re-check cadence per data source (CGHS, PMJAY, MJPJAY, NPPA) to Role 5, since these move at different rates.

14. Privacy & Security

Honest answer: Correctly scoped in principle (uploaded documents only, no unnecessary long-term storage) — but "avoid long-term storage unless required" needs to become an actual policy, not a soft intention.

Fix:

- Write a concrete **data retention rule**: e.g., auto-delete uploaded documents and extracted case data after N days unless the user explicitly opts to keep their case history.
- Encrypt at rest and in transit (Postgres encryption + HTTPS — standard, not exotic).
- Your Kadi **consent boundary** (§4.5 — per-case opt-in before BillNyay's findings feed SchemeSetu/DawaCheck) is already a strong, specific answer to "how do you protect sensitive data" — lead with that in viva, it shows the team thought about cross-module data flow, not just per-module storage.

15. Scalability

Honest answer: Dockerized + modular is a reasonable *architecture* answer, but "thousands of users simultaneously" is not something you need to prove for an FYP.

Fix — reframe the target, don't over-engineer:

- Your real target is a **working demo + a real pilot cohort** (tens to low hundreds of users), not production-internet scale. Say that explicitly rather than implying you're scale-tested for thousands — an examiner will respect an honest, correctly-scoped claim far more than an inflated one that falls apart under a follow-up question.
- If you want a genuine scalability story for the report: basic load-testing (e.g., Locust) against a target of "50 concurrent users" is achievable and honest; "thousands" is not, and you shouldn't claim it.

16. AI vs Professional Advice

Honest answer: Real risk of over-reliance, and your scope statement ("decision-support system," informational not medical/legal advice) is the correct position.

Fix — make the disclaimer functional, not just written:

- A disclaimer banner **the user must actively acknowledge** before generating an appeal or viewing scheme guidance (not buried in a footer) is a stronger, more honest design than passive text — and it's a good answer if a viva panel asks "how do you prevent misuse."
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Pattern across all 16

Notice most of these don't need new engineering — they need **honest labeling, visible timestamps/confidence scores, and explicit disclaimers** at the right points in the UI and output documents. That's the cheapest, most defensible fix available to you, and it's also exactly what turns "weaknesses a teammate found" into "known limitations we designed around" in your final report — which is a much stronger story for both your guide and any external examiner.